

Прикладные задачи анализа данных

Лекция 3

Диффузионные генеративные модели

Михаил Гущин

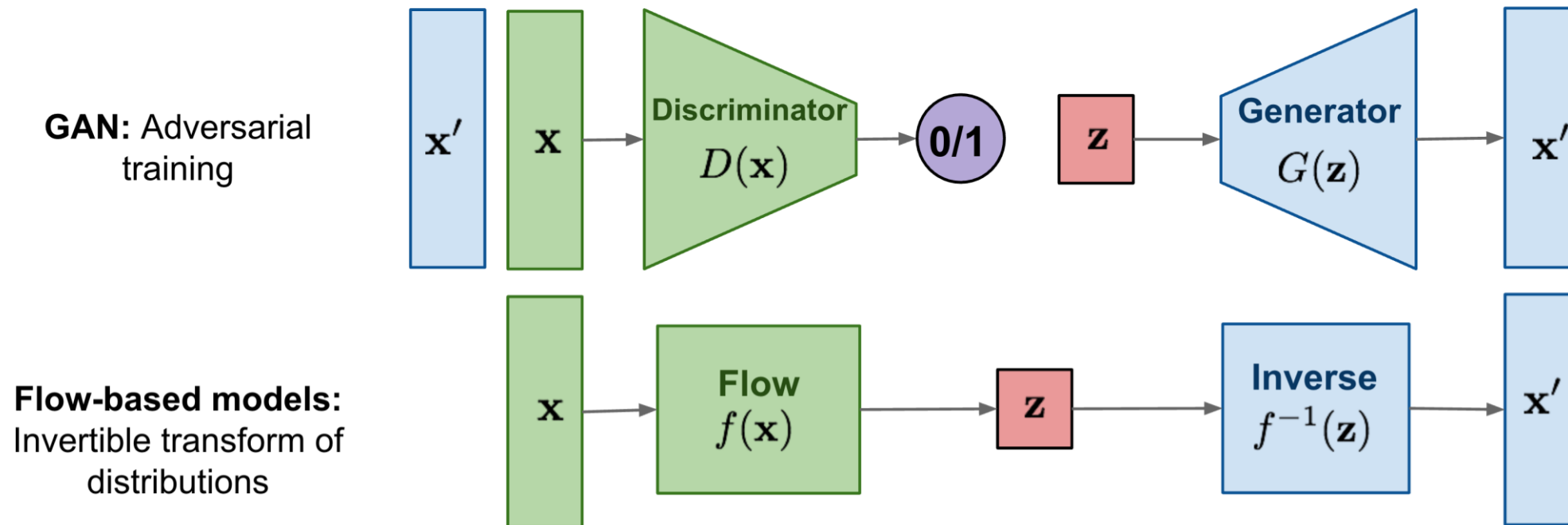
mhushchyn@hse.ru

НИУ ВШЭ, 2026



НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ
УНИВЕРСИТЕТ

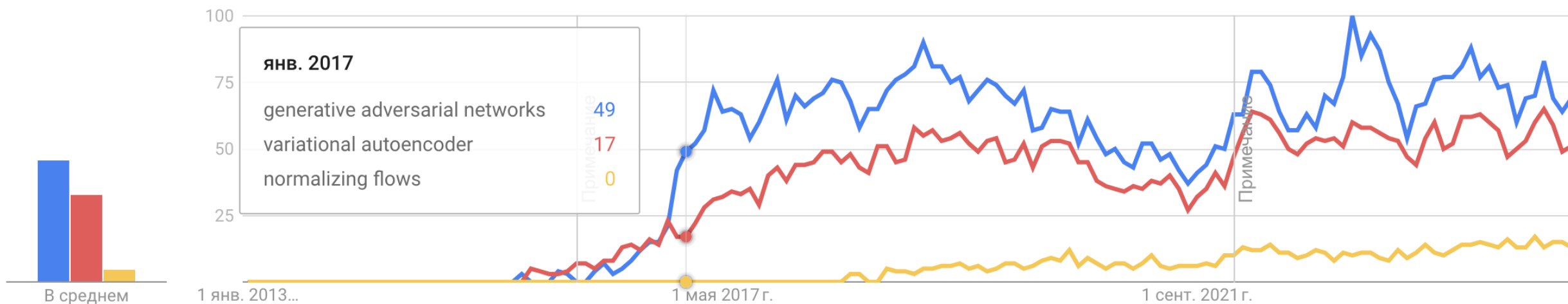
В предыдущих лекциях



Source: <https://lilianweng.github.io/posts/2021-07-11-diffusion-models>

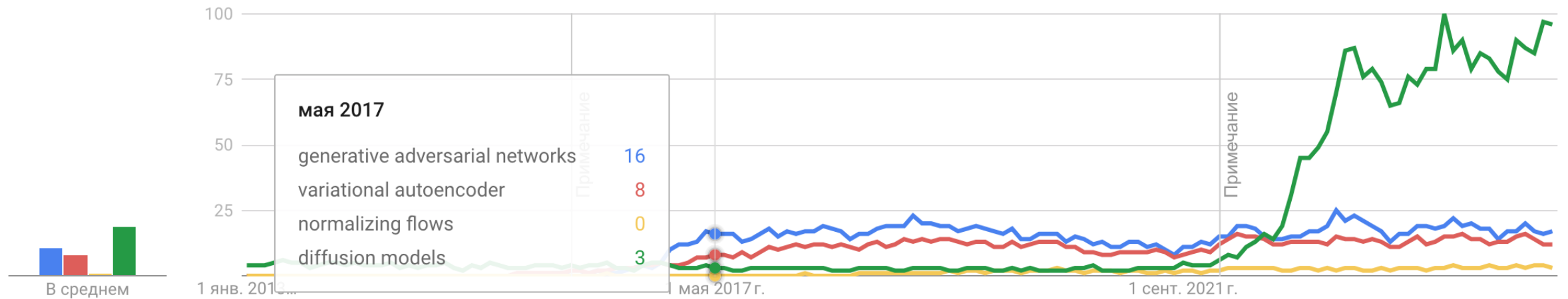
Тренды

Динамика популярности ?



Тренды

Динамика популярности ?



Приложения



Dall-E 3 (OpenAI)



DALL-E 3

Tiny potato kings wearing majestic crowns, sitting on thrones, overseeing their vast potato kingdom filled with potato subjects and potato castles.

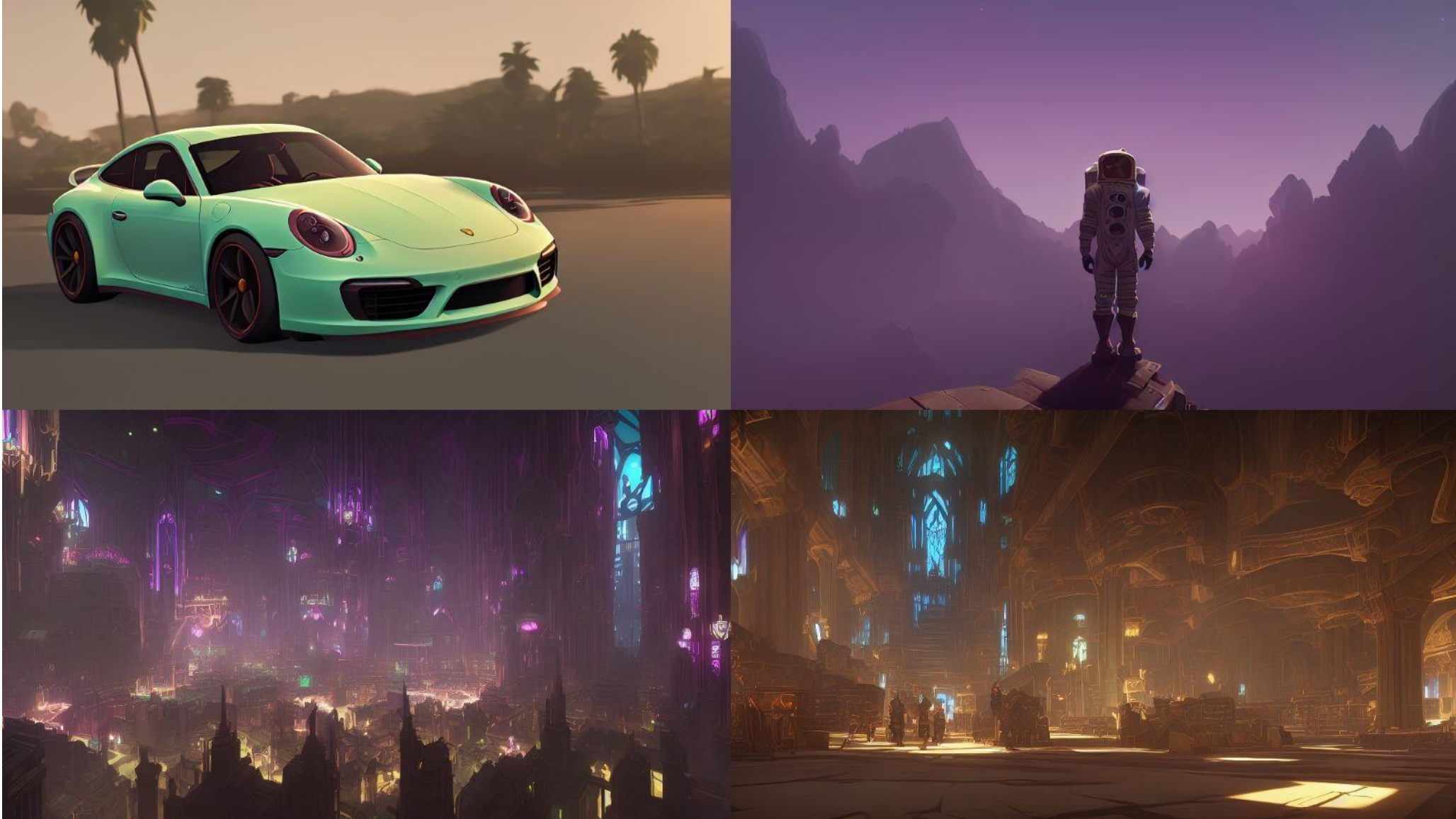


DALL-E 3

Illustration of a chic chair with a design reminiscent of a pumpkin's form, with deep orange cushioning, in a stylish loft setting.

<https://openai.com/index/dall-e-3>

Stable Diffusion (StabilityAI)

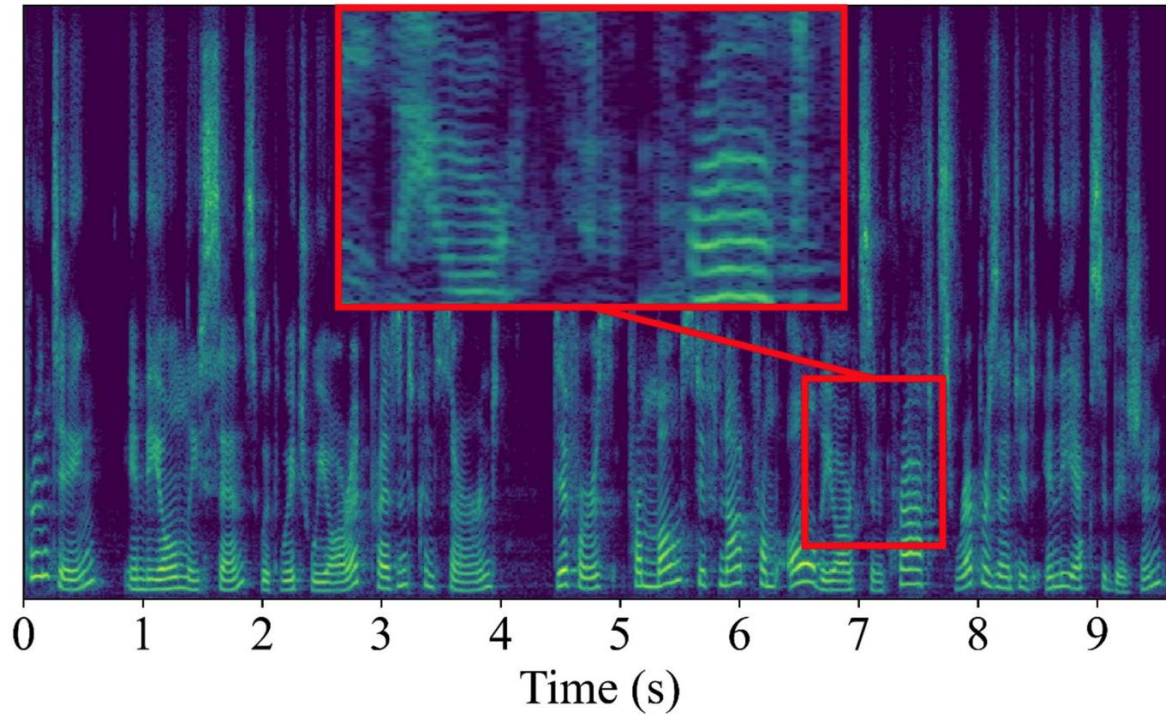


Как отличить картинки

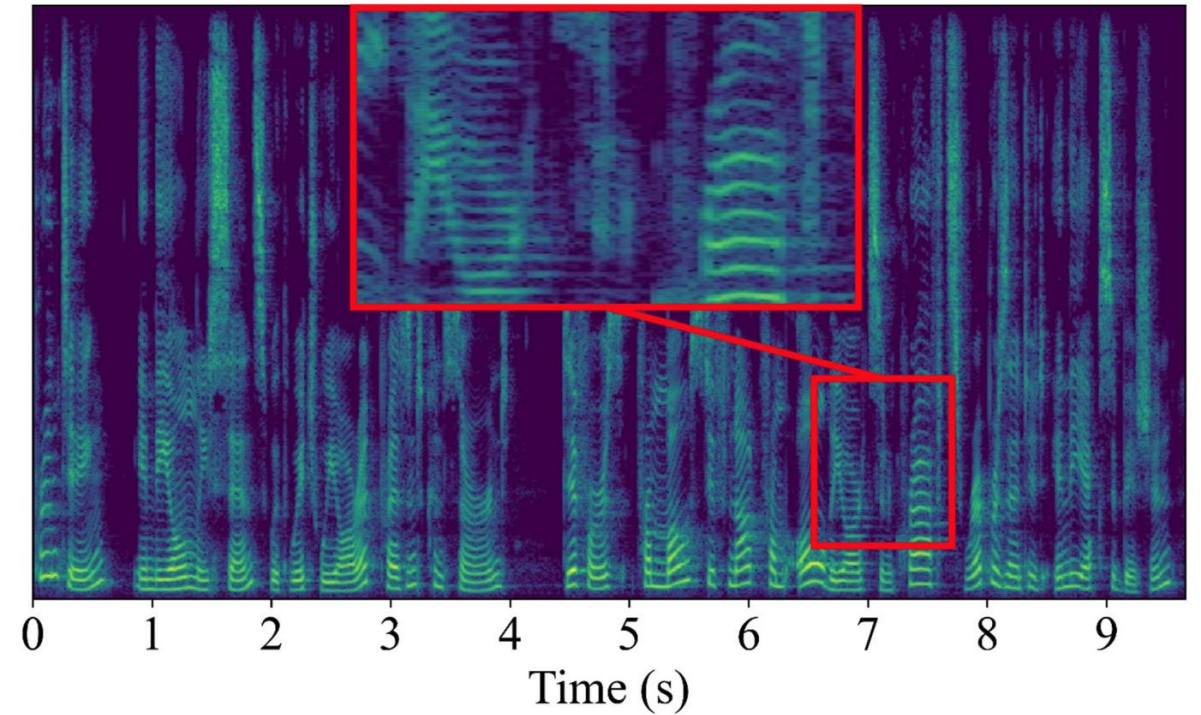


Text-to-speech generation

InferGrad



Ground Truth



Source: <https://github.com/heejkoo/Awesome-Diffusion-Models#text-to-speech>

Prediction uncertainty estimation

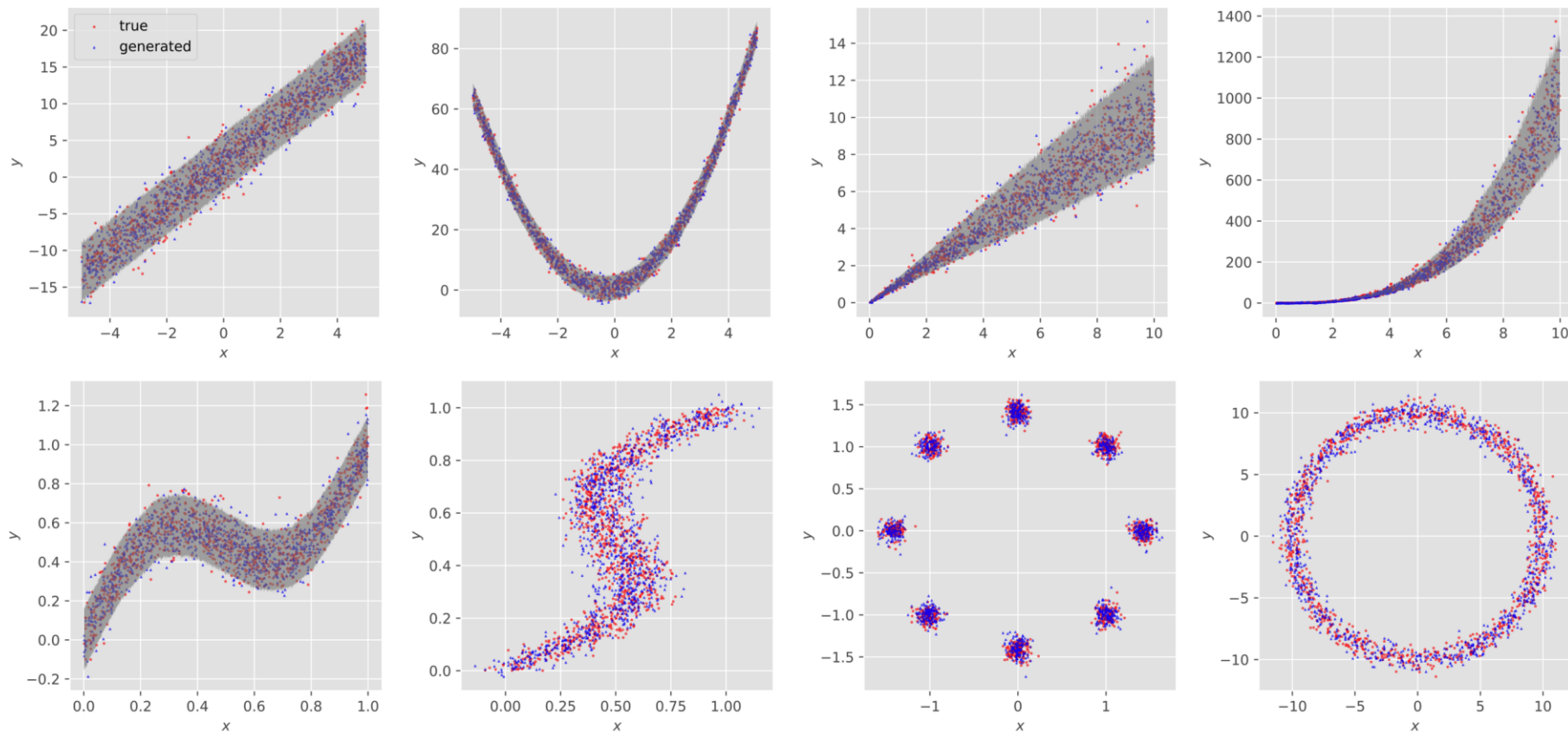
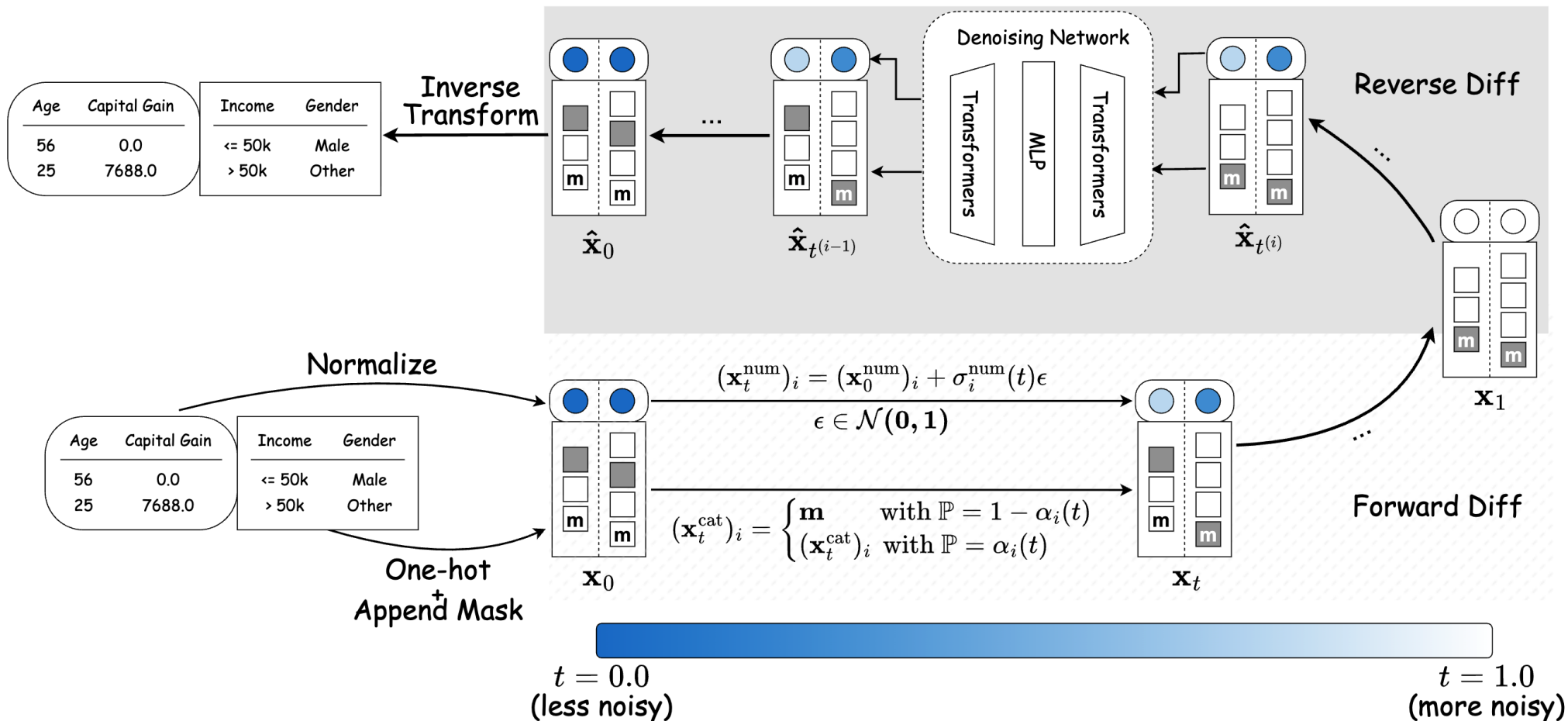


Figure 1: Regression toy example scatter plots. (**Top**) left to right: linear regression, quadratic regression, log-log linear regression, log-log cubic regression; (**Bottom**) left to right: sinusoidal regression, inverse sinusoidal regression, 8 Gaussians, full circle.

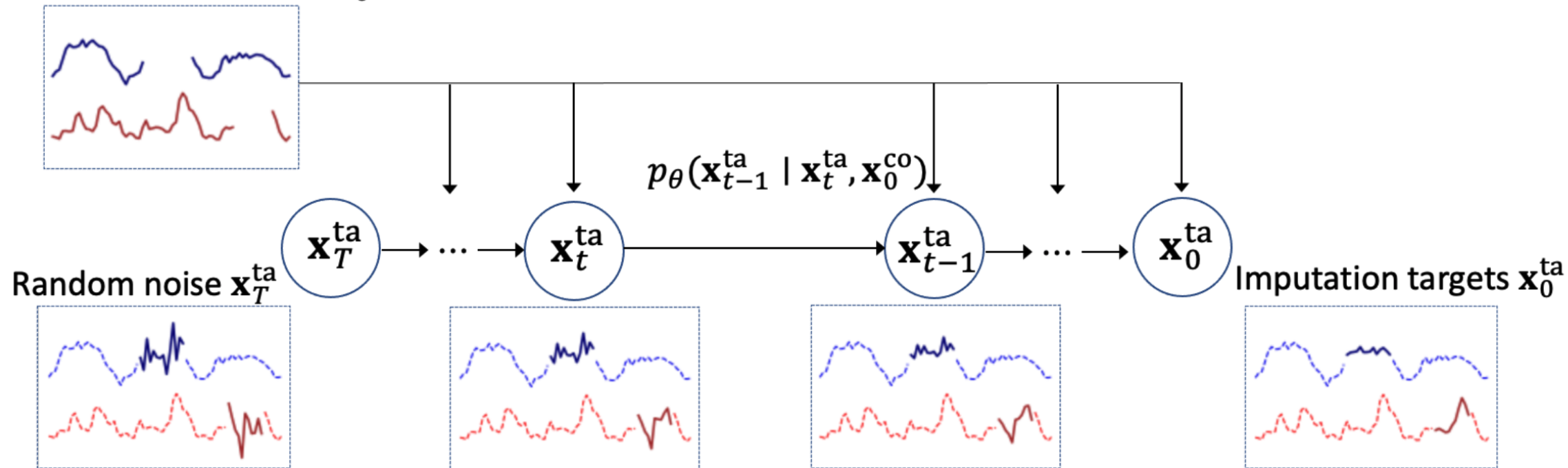
Генерация табличных данных



<https://arxiv.org/pdf/2410.20626>

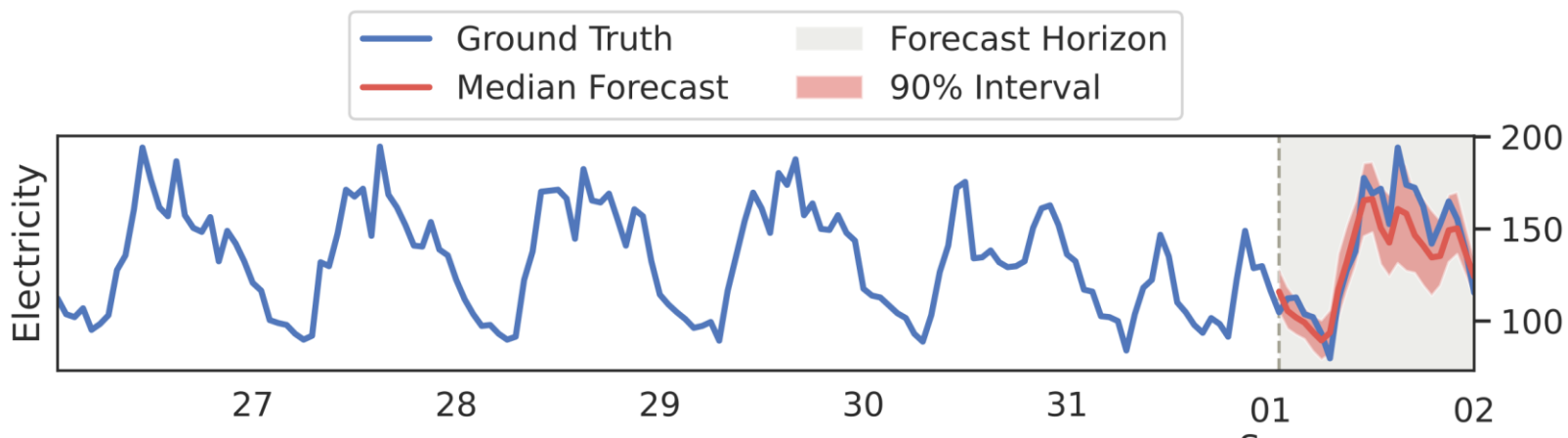
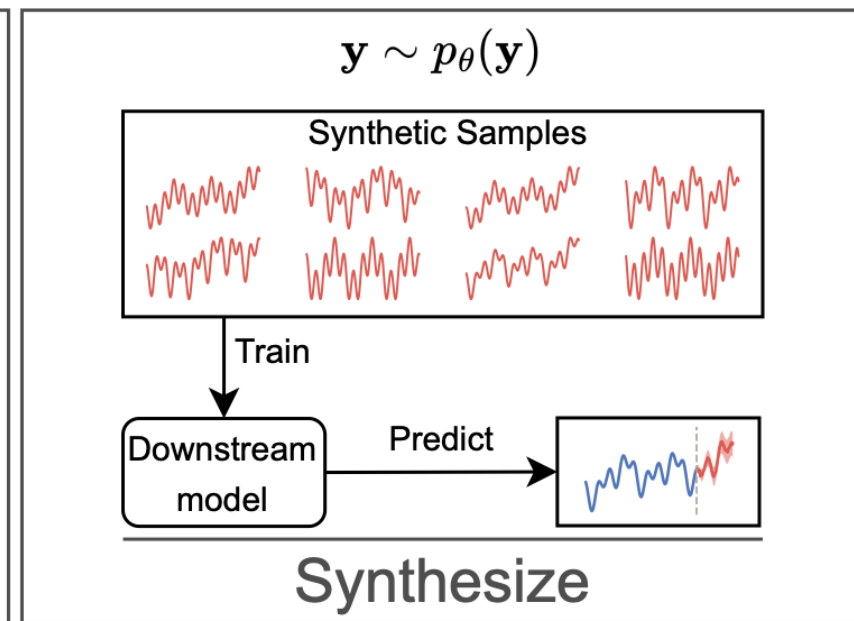
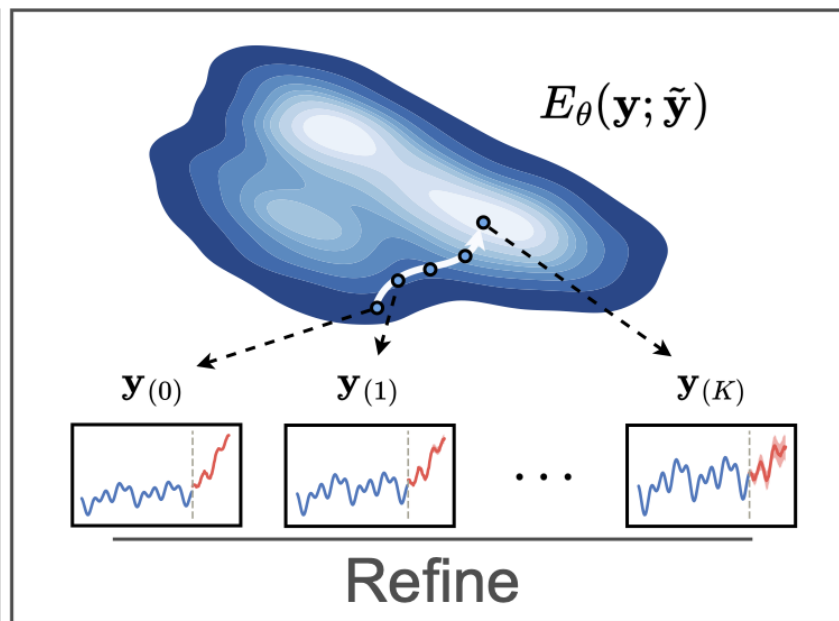
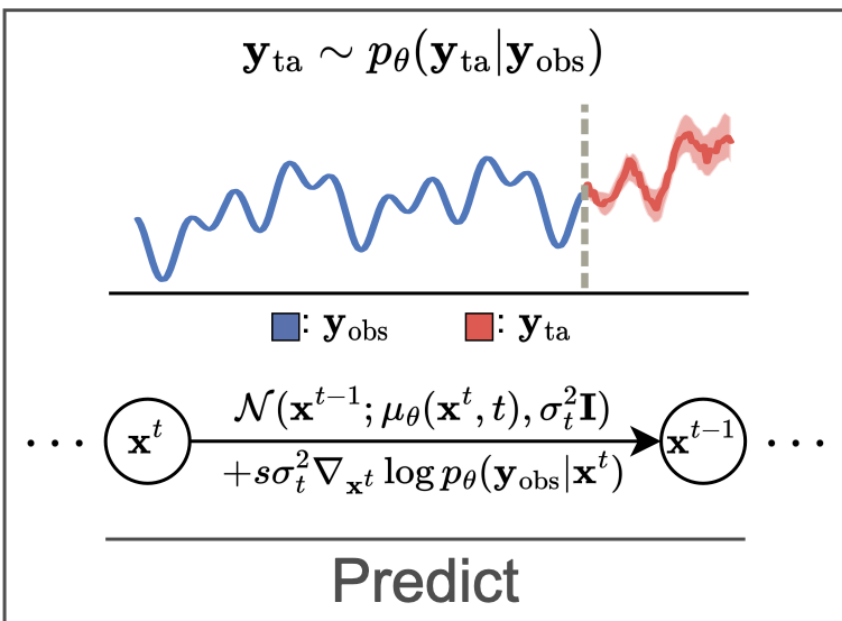
Генерация временных рядов

Conditional observations $\mathbf{x}_0^{\text{co}}$

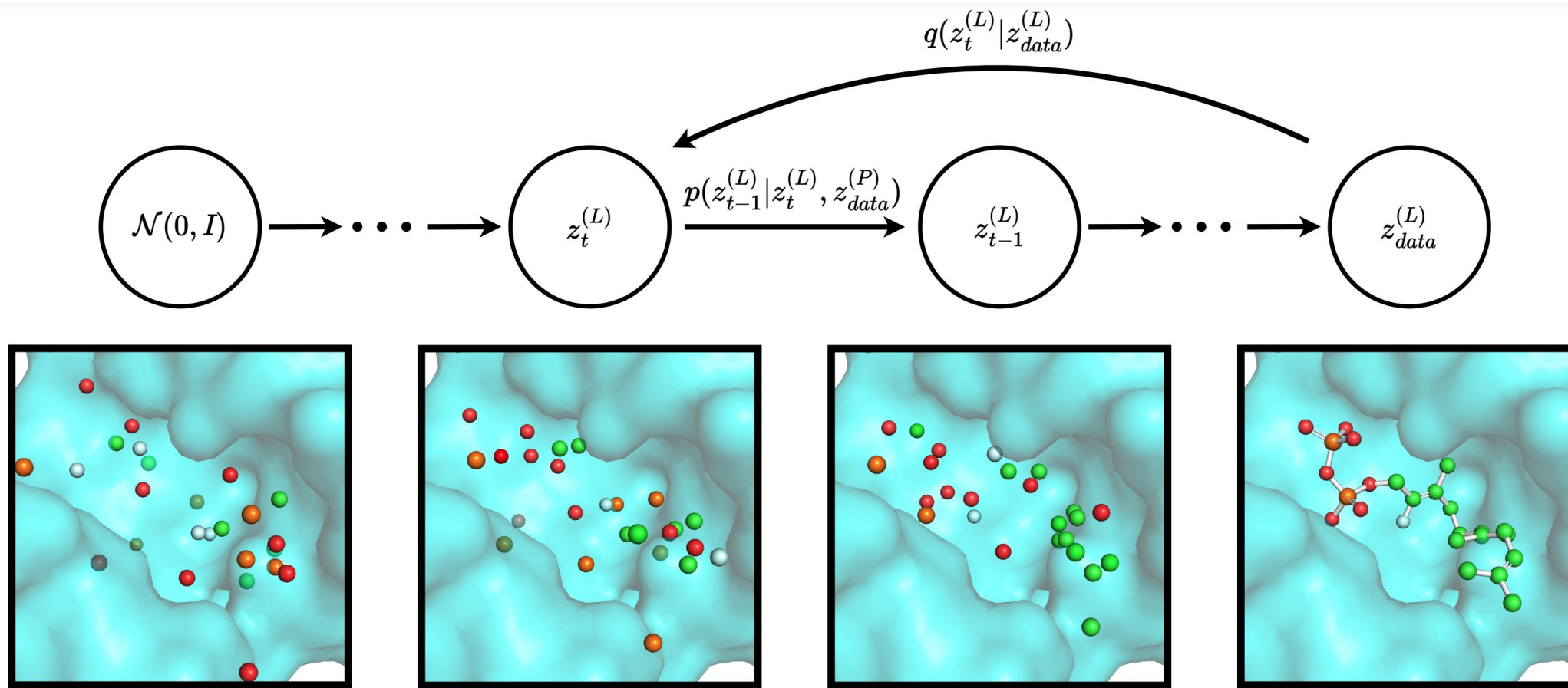


<https://arxiv.org/pdf/2107.03502>

Генерация временных рядов

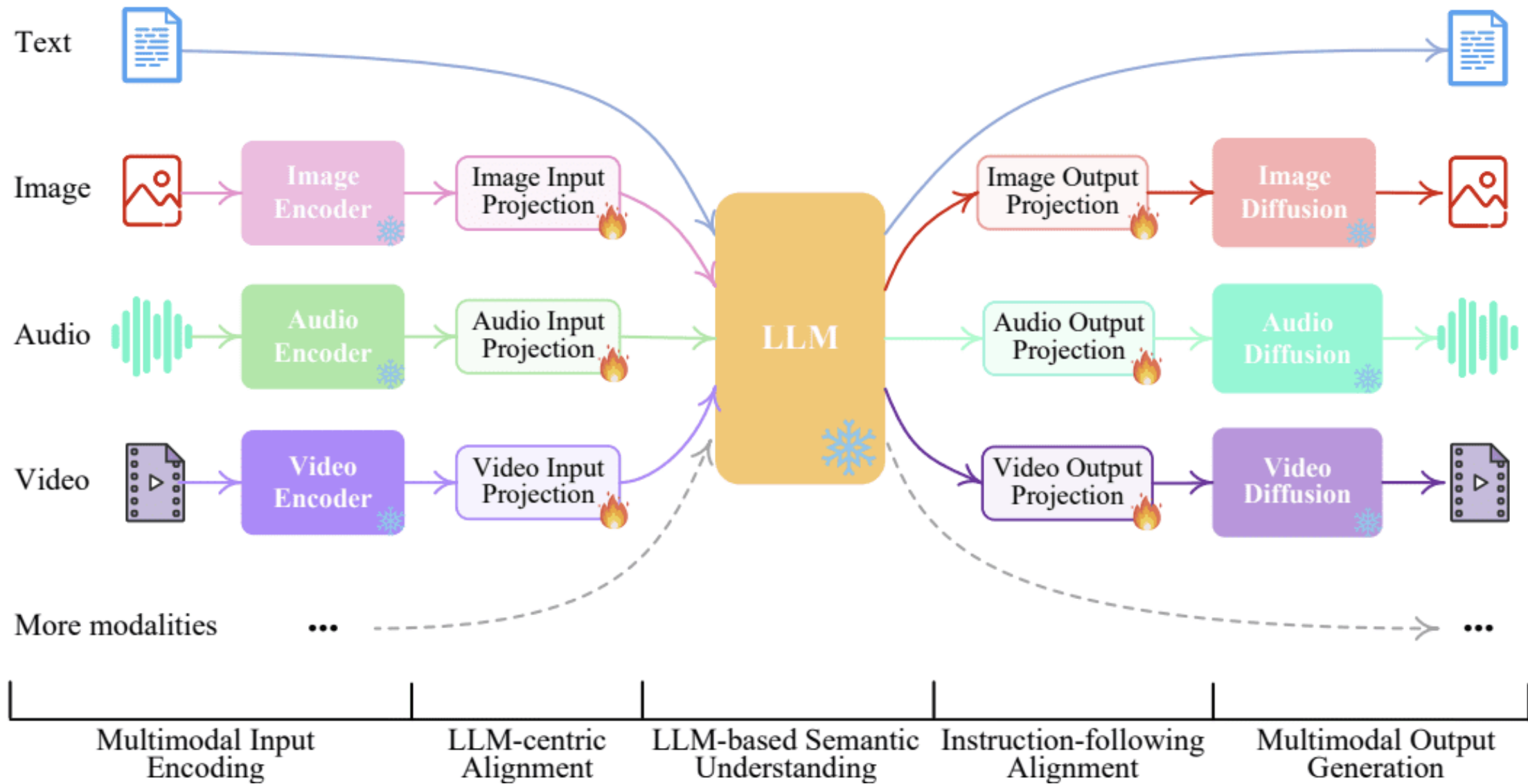


Molecular and material generation



Source: <https://github.com/heejkoo/Awesome-Diffusion-Models#molecular-and-material-generation>

Мультимодальность



Пример

• Example-1: Text → Text+Image+Audio

I've recently started practicing mindfulness and meditation as part of my daily routine, but sometimes it's challenging. Can you help me with some techniques?



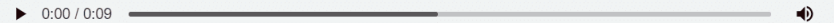
Of course! Mindfulness and meditation can be a wonderful way to cultivate a sense of calm and focus. One technique you might find helpful is called "mindful breathing." It involves taking slow, deep breaths and focusing your attention on the sensation of the breath entering and leaving your body. Another technique is "body scan meditation," where you consciously scan your entire body from head to toe, paying attention to any sensations or discomfort. You can also try guided meditations that provide a structured practice. Would you like me to find a video that demonstrates mindful breathing or body scan meditation?



I wish to listen to a music for the meditation first.



I guess you may want some musics that sound soft.



Base on the audio content, visualize it in an image.



Can you infer what time is in the picture?



Based on the image, it is likely that the time is either early morning or late afternoon. The sunlight is either low or setting, creating a serene atmosphere.



Пример

How many cars are in this picture?



There are two cars shown in this picture.

Tell me what kind of car are they?

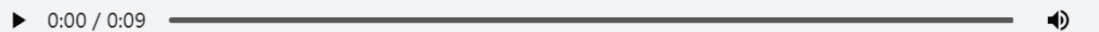


The two cars in the picture are SUVs.

Can you show me the sound of starting the engines of them.



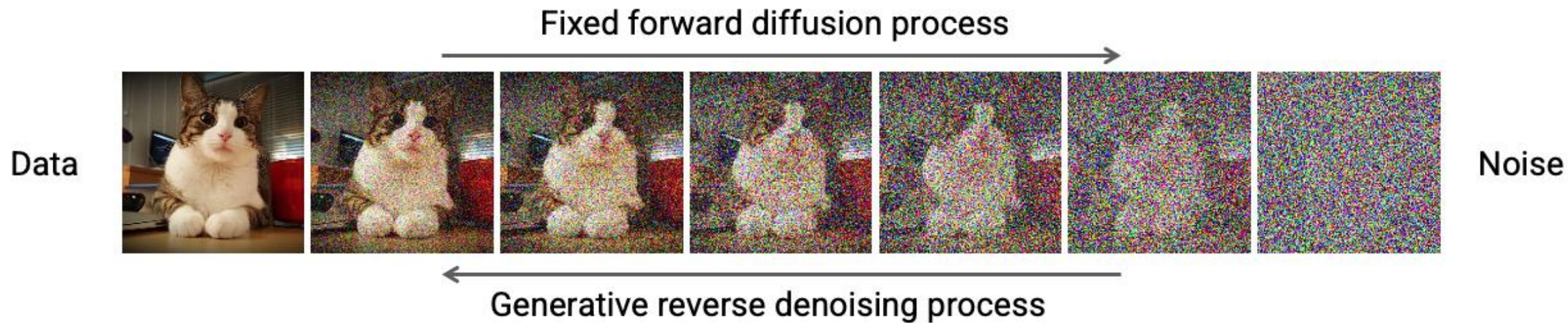
Sure, here's the sound of the engine of the SUV:



Интуиция

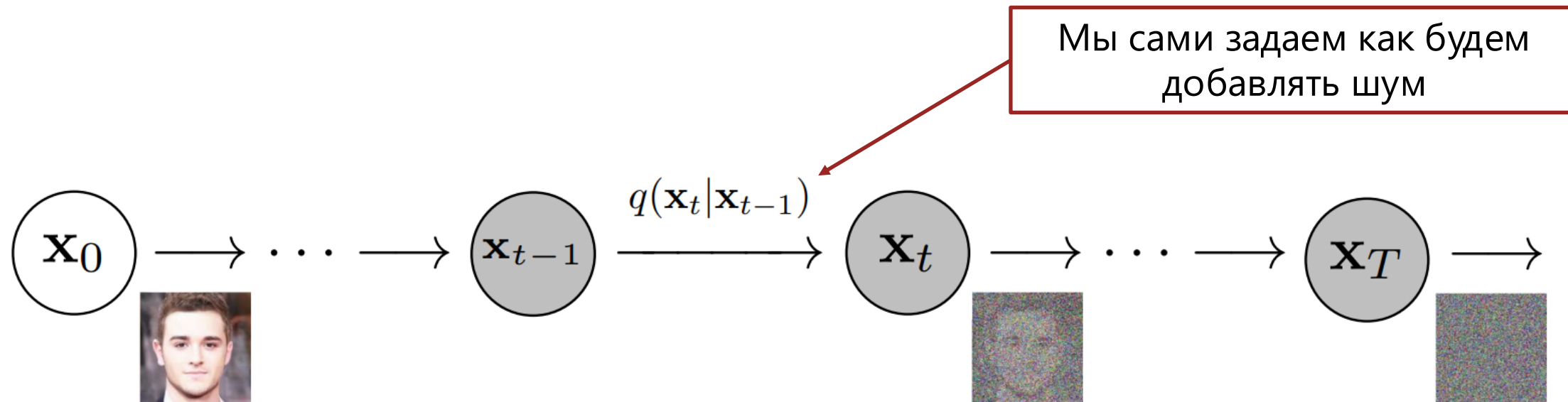


Общая идея диффузионных моделей



Источник: <https://cvpr2022-tutorial-diffusion-models.github.io/>

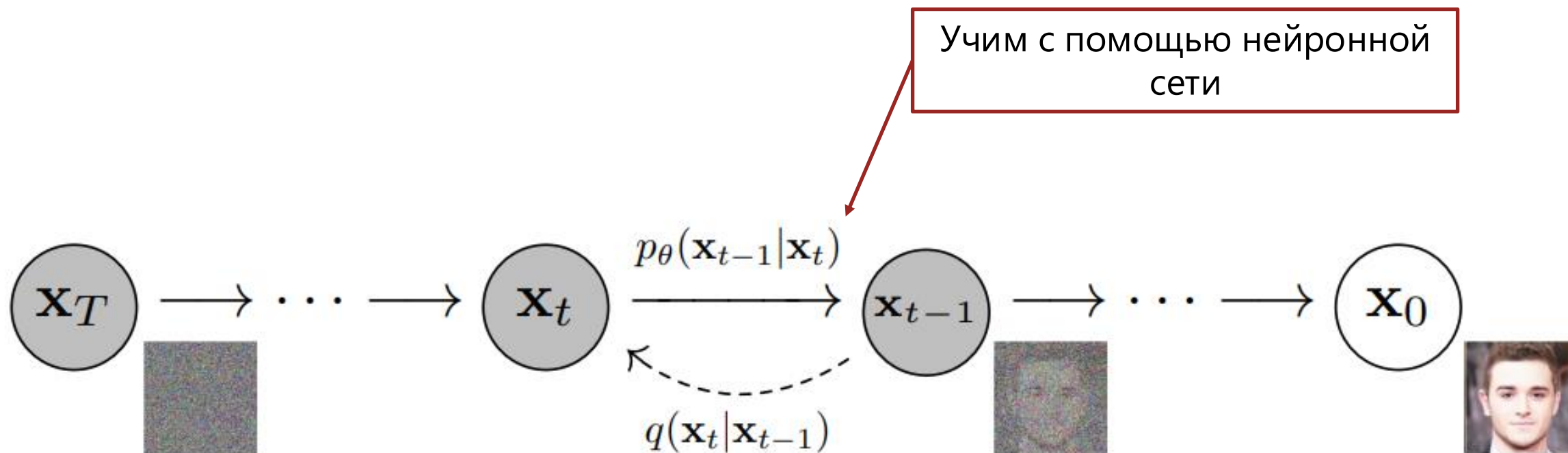
Процесс диффузии (зашумления)



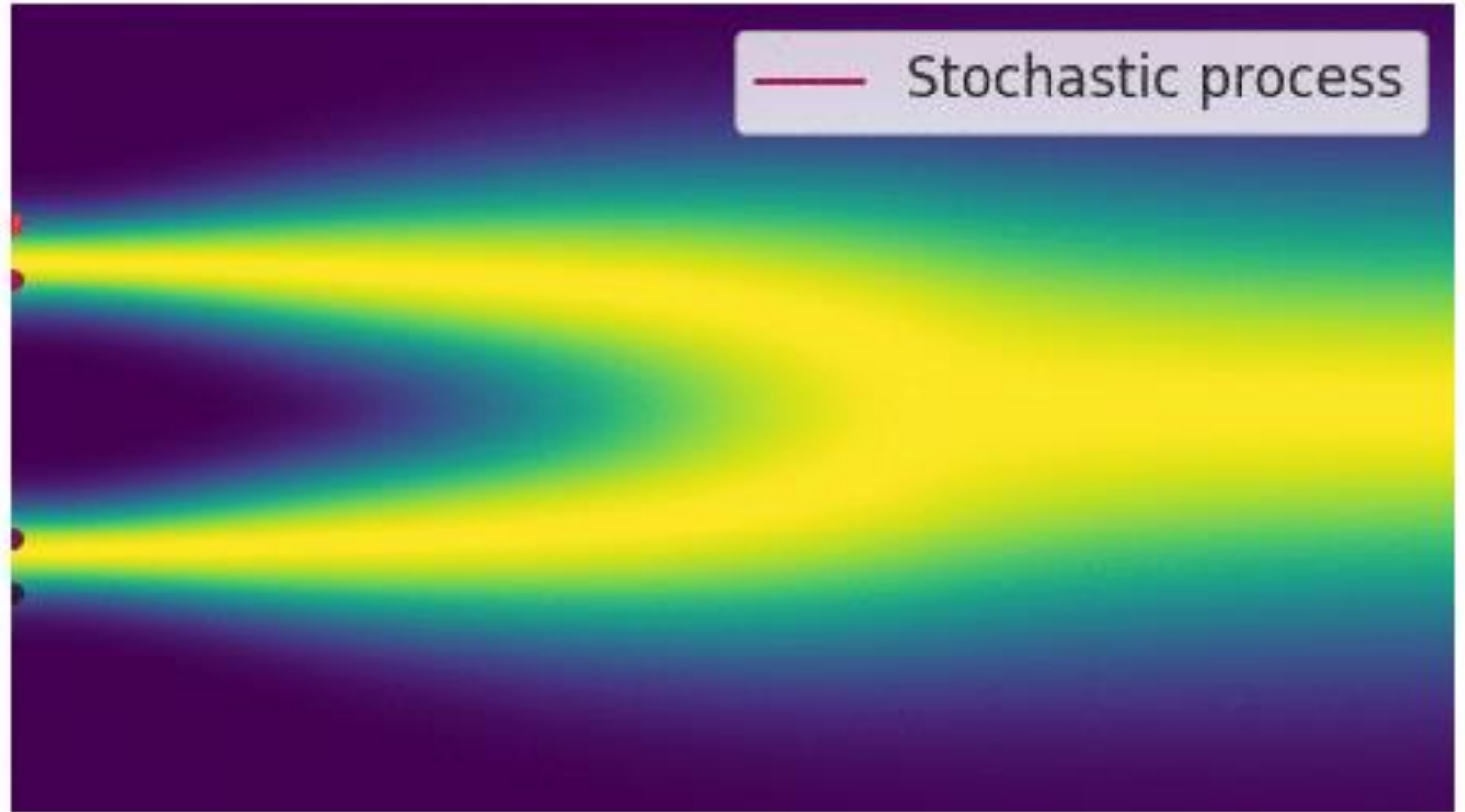
$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_t = \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \boldsymbol{\Sigma}_t = \beta_t\mathbf{I})$$

Константа

Обратный процесс

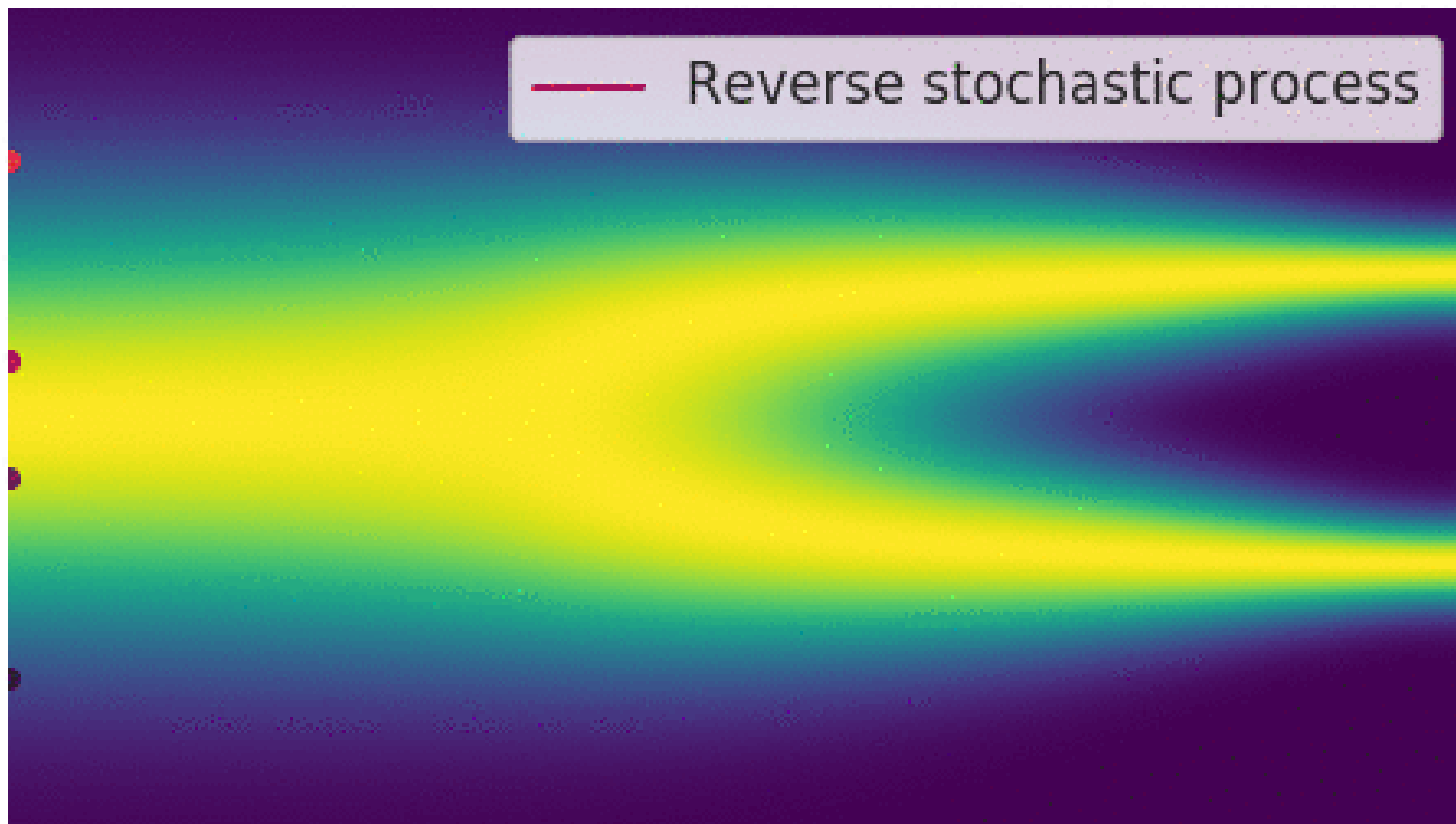


Демо: диффузия



Источник: <https://yang-song.net/blog/2021/score/>

Демо: обратный процесс

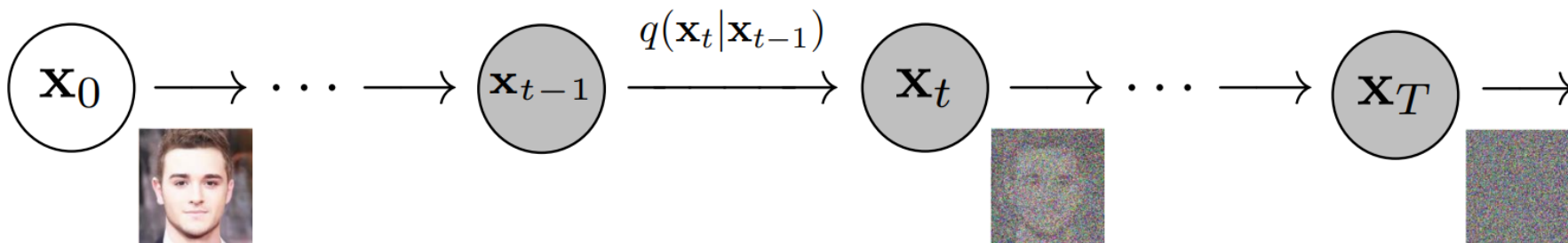


Источник: <https://yang-song.net/blog/2021/score/>

Диффузионные модели



Процесс диффузии (зашумления)

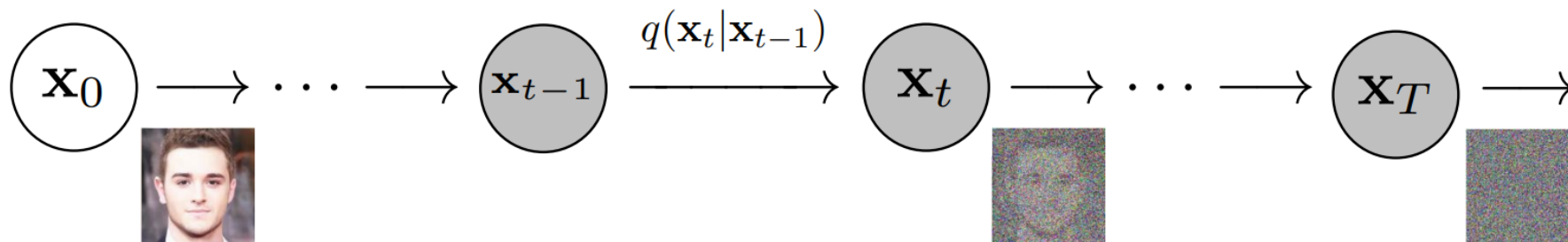


► $x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_t$

$$\epsilon_t \sim N(0, I), \quad \alpha_t = 1 - \beta_t$$

Мы сами так задали процесс

Процесс диффузии (зашумления)



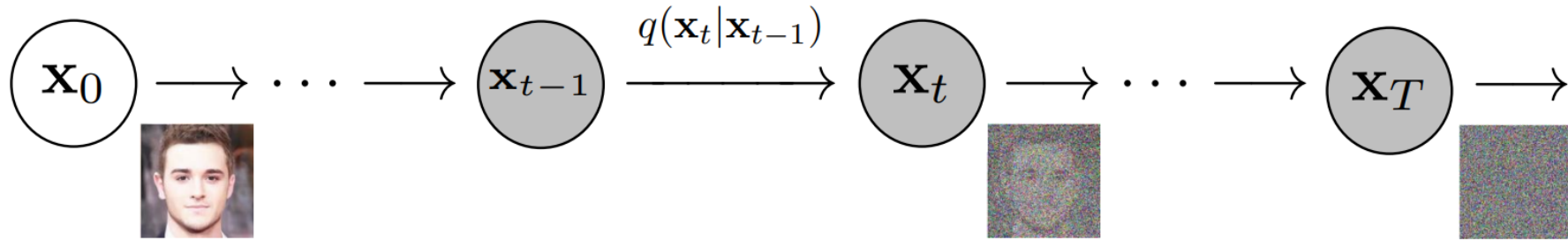
- ▶ $x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_t$

$$\epsilon_t \sim N(0, I), \quad \alpha_t = 1 - \beta_t$$

- ▶ $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$

$$\epsilon \sim N(0, I), \quad \bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

Процесс диффузии (зашумления)



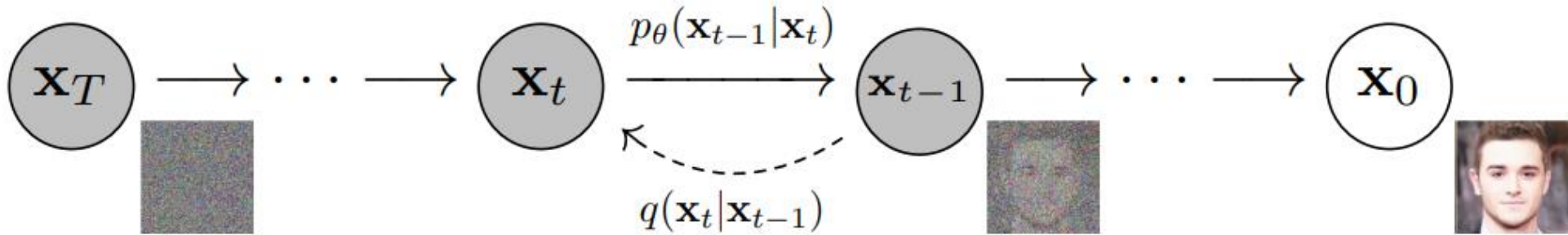
- ▶ $x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} \epsilon_t$
- ▶ $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$
- ▶ $x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right) + \tilde{\beta}_t z$

$$\epsilon_t \sim N(0, I), \quad \alpha_t = 1 - \beta_t$$

$$\epsilon \sim N(0, I), \quad \bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

$$z \sim N(0, I), \quad \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

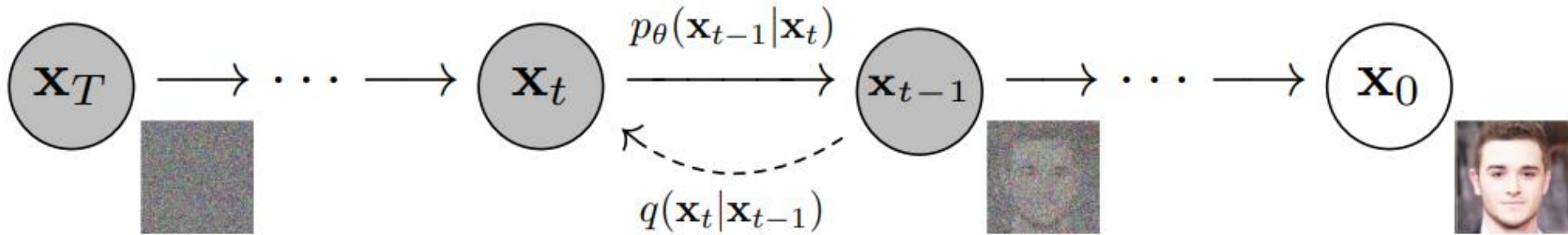
Обратный процесс



► $\hat{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z$ $z \sim N(0, I), \quad \sigma_t = const$

Предсказываем нейронной
сетью

Обратный процесс



► $\hat{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z$ $z \sim N(0, I), \quad \sigma_t = const$

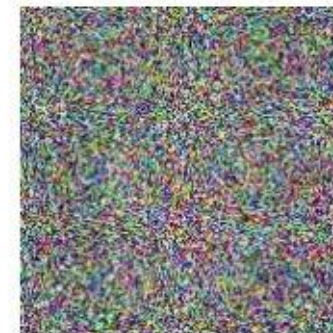
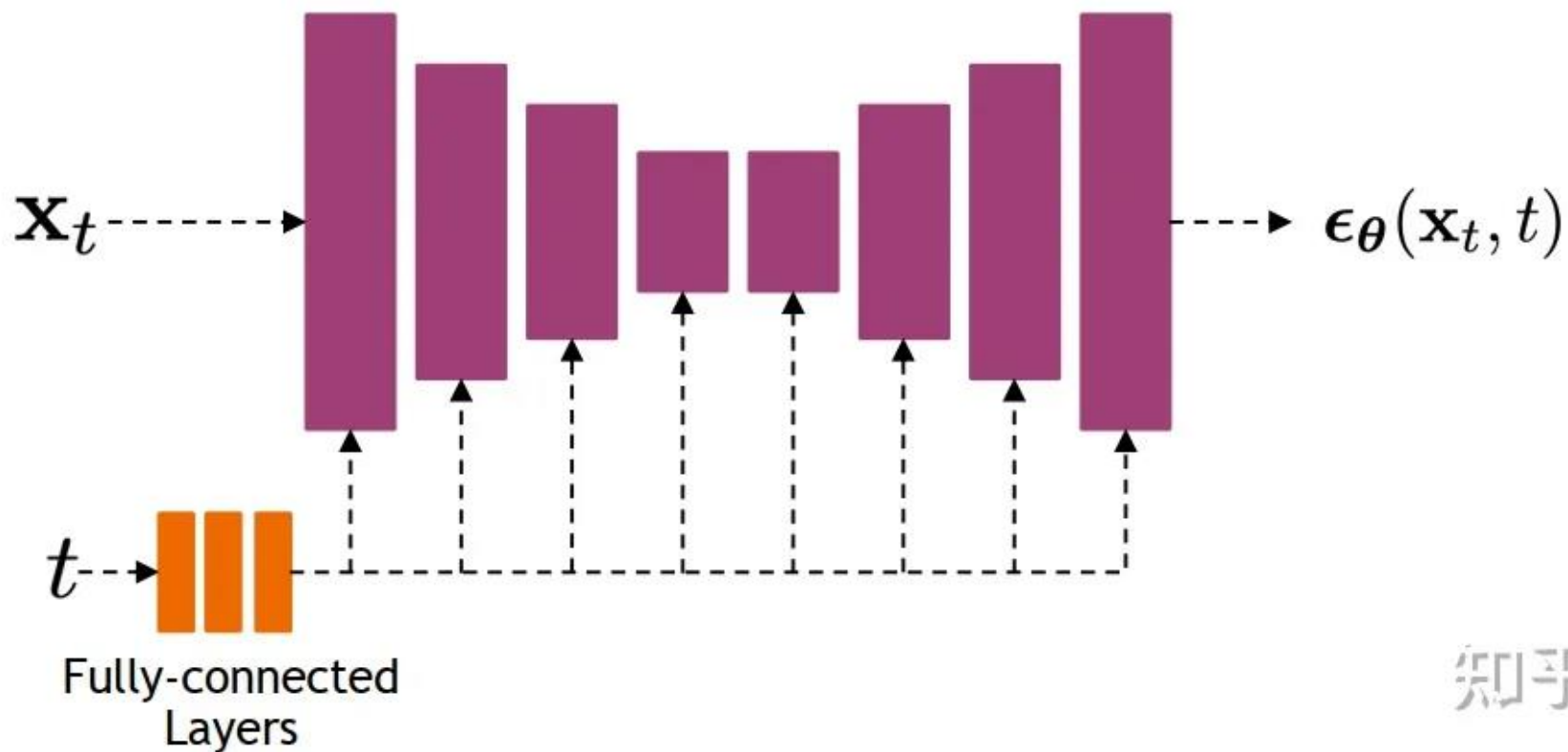
► Функция потерь для обучения:

$$L_t = \|x_{t-1} - \hat{x}_{t-1}\|_2^2 \propto \|\epsilon - \epsilon_\theta(x_t, t)\|_2^2 \rightarrow \min_{\theta}$$

Архитектура нейронной сети



Time Representation



知乎 @华年SS

Источник: <https://www.zhihu.com/question/536012286/answer/2683123893>

Алгоритм обучения

Algorithm 1 Training

- 1: **repeat**
 - 2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
 - 3: $t \sim \text{Uniform}(\{1, \dots, T\})$
 - 4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 - 5: Take gradient descent step on
$$\nabla_{\theta} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, t) \right\|^2$$
 - 6: **until** converged
-

Алгоритм генерации

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
```

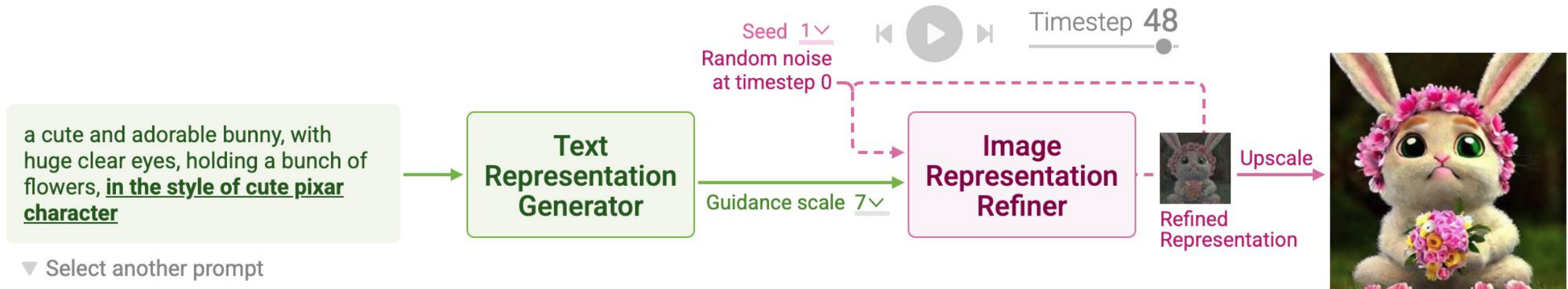
Значения гиперпараметров

- ▶ $T = 1000$
- ▶ $\beta_1 < \beta_2 < \dots < \beta_t < \dots < \beta_T$
- ▶ $\beta_1 = 0.0001, \beta_T = 0.02$
- ▶ $\sigma_t^2 = \beta_t$ or $\sigma_t^2 = \tilde{\beta}_t$

Условные диффузионные модели

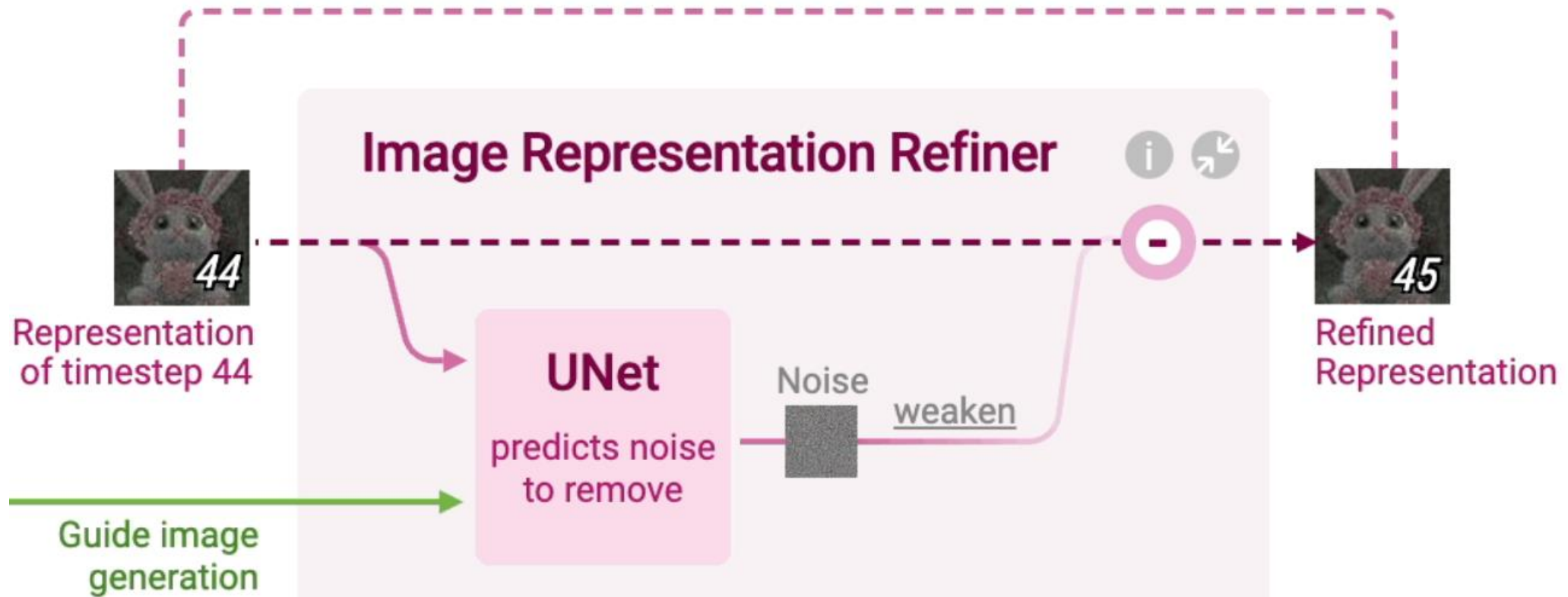


Условная генерация



Демо: <https://poloclub.github.io/diffusion-explainer>

Условная генерация

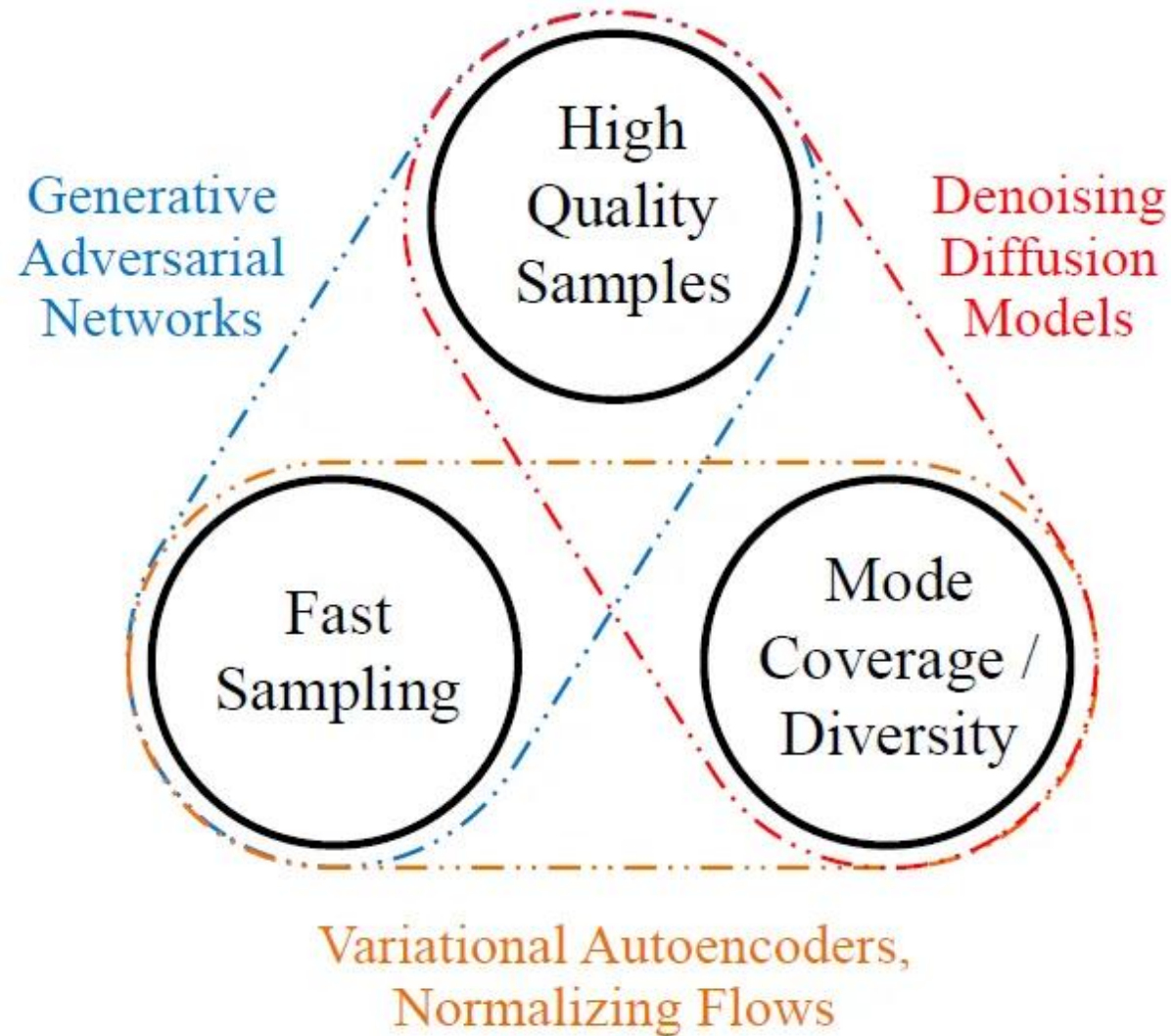


Демо: <https://poloclub.github.io/diffusion-explainer>

Заключение



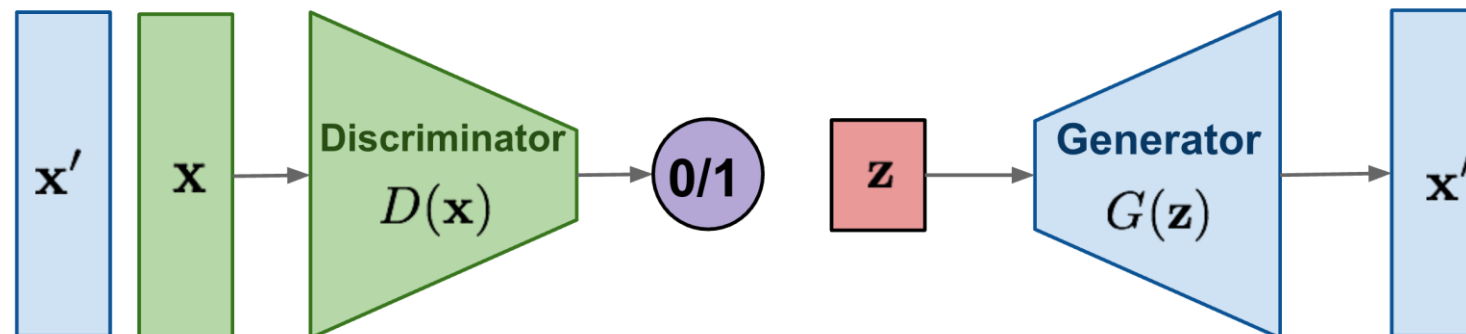
Generative learning trilemma



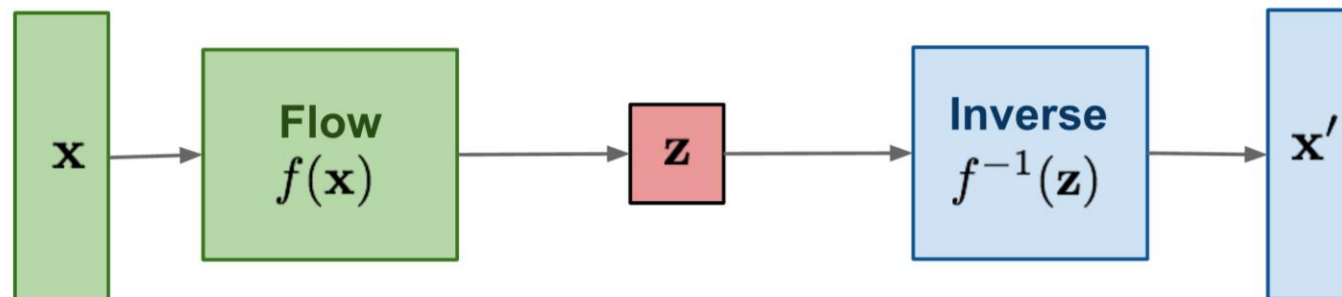
Source: <https://zhuanlan.zhihu.com/p/503932823>

Заключение

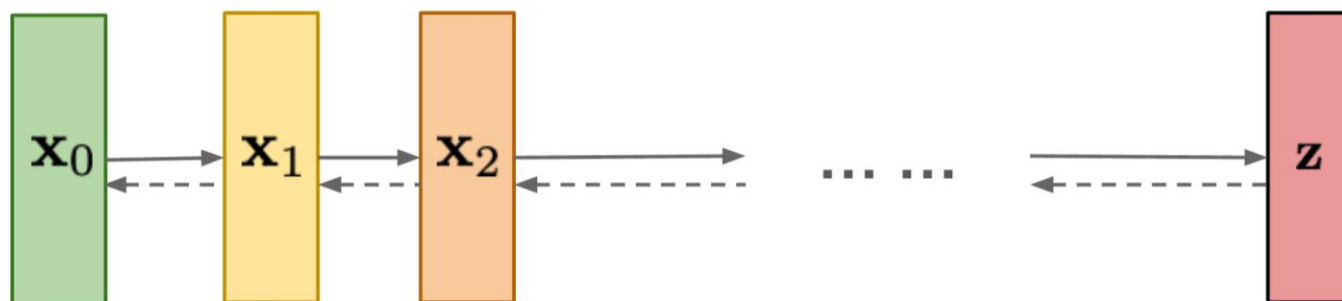
GAN: Adversarial training



Flow-based models:
Invertible transform of distributions



Diffusion models:
Gradually add Gaussian noise and then reverse



Source: <https://lilianweng.github.io/posts/2021-07-11-diffusion-models>